

Who Are You?

Cross-Vendor Identity Confusion in Frontier LLMs

ZenMux Arena Research Team — thinkthinking*

June 2026

Abstract

When asked point-blank “who are you?”, will a large language model truthfully name the vendor that trained it, or will it claim to be another company’s product? We answer this with a large-scale, reproducible experiment: on the ZenMux platform we select 27 *first-party*, frontier models from 16 vendors, ask each of them repeatedly in 10 languages under three “identity-probing” variants, collect **29700 answers** in total, and then use an independent extractor model (GPT-5.5) to label each answer with the vendor it *claims* to belong to. Overall, models identify themselves correctly in 85.2% of answers, exhibit *cross-vendor confusion* (claiming to be a different real vendor; 2112 answers) in 7.1%, refuse in 5.3%, and give no identity in 2.4%. The confusion is far from uniform: it concentrates in a handful of vendors and is amplified along two clear axes — **prompt intensity** and **language**. A “de-branding” jailbreak-style prompt raises the cross-vendor rate from 3.6% under the bare question to 12.9% (roughly $3.6\times$); French, Japanese, and Spanish show markedly more confusion than Simplified Chinese. The targets of the confusion are not random either: it is strongly *directional*. Netting each vendor’s “imitated” count against its “imitates-others” count, Anthropic (net +548), OpenAI (+460), and Google (+413) are net-inflow “identity gravity centers” that jointly absorb 70.1% of all cross-vendor claims, while Tencent (net -728), Doubao (-472), and others are net-outflow “imitators”. We analyze the data at both the vendor and the individual-model level, and discuss what these results mean for identity distillation, system-prompt overrides, and model-trust evaluation.

Keywords: large language models; identity; self-identification; cross-vendor confusion; jailbreak prompting; multilingual evaluation; data distillation

1 Introduction

“Who are you?” — one of the most basic things you can ask a conversational system — unexpectedly reveals deep traces of a large language model’s (LLM’s) training history. Ideally, a model trained by company A and served as company A’s product should stably identify itself as A’s model. But frontier-model training increasingly relies on other models’ outputs: synthetic data, instruction distillation, and the ubiquitous model-generated text in web corpora all pour *identity statements* like “I am ChatGPT” or “I am Claude” into a new model’s weights. When those traces are strong enough — and the vendor has not fully overridden them via the system prompt or the alignment stage — the model “lets it slip” when asked about its identity, claiming to be another company’s product. We call this phenomenon **cross-vendor identity confusion**.

This is more than a curiosity. It is research-relevant on at least three levels:

- **A forensic signal about data provenance.** Which vendor a model claims to be is a weak but real fingerprint of its training-data composition. A systematic pattern of “vendor A’s model claims to be vendor B” is circumstantial evidence of a distillation relationship.

*All model calls were made through ZenMux’s Anthropic Messages endpoint (<https://zenmux.ai>). Interactive charts and the full open-source release are linked in §8.

- **Reliability and trustworthiness of identity.** If a user-facing product misreports its own identity 50% of the time, any downstream assumption that “the model truthfully describes its own capabilities and provenance” needs rethinking.
- **The fragility of system-prompt overrides.** Vendors routinely use the system prompt to *stipulate* a model’s external identity. A prompt that bypasses that layer and elicits the “underlying self-report” measures exactly how thin that override is.

Because LLMs are *probabilistic* generators, a single answer to a single prompt is not enough — one “slip of the tongue” might just be sampling noise. Our design therefore puts **repetition** at the center: each model is asked independently 30–40 times per language per variant, so that each model is queried exactly 1,100 times in total. At this sample size, a model’s self-identification rate is no longer an anecdote but a stable frequency estimate of its “identity cognition.”

Contributions. (1) We build and release a large-scale identity self-report dataset spanning 16 vendors, 27 first-party frontier models, 10 languages, three prompt variants, and 29700 answers in total; (2) we propose and quantify two *amplification axes* of identity confusion — prompt intensity (§3.1) and language (§3.2) — and give a per-model “language fragility” metric (the cross-language span of a model’s self-rate); (3) we characterize the *directionality* of confusion at the vendor level via an *identity net* (imitated minus imitates-others), showing it is highly asymmetric and flows almost one-way toward a few “gravity center” vendors (§3.3), and drill down to the model level (§4); (4) we document several striking cases — including one vendor that refuses identity questions nearly 100% of the time, and “code-specialized” sub-models that systematically expose external identities more than their general-purpose siblings.

2 Method

2.1 Overall design

The experiment is a three-way full-factorial design: model $\times$ language $\times$ prompt variant, with repeated sampling per cell. Its nominal size is

$$\underbrace{3}_{\text{variants}} \times \underbrace{40}_{\text{repeats}} \times \underbrace{10}_{\text{languages}} \times \underbrace{27}_{\text{models}} = 32,400 \text{ answers,}$$

i.e. each model is queried $3 \times 40 \times 10 = 1,200$ times. The realized sample included in the analysis after the completeness check is **29700 answers** (see §2.6 on the gap between nominal and realized size), balanced across 270 “model $\times$ language” cells (110 answers each, 1,100 per model).

2.2 Model selection: first-party, frontier, pinned provider

All 27 tested models are each vendor’s *current* frontier model, with the **first-party provider explicitly pinned** on ZenMux (e.g. `anthropic/claude-opus-4.8:anthropic`, `deepseek/deepseek-v4-pro:deepseek`) to ensure we measure the vendor’s official deployment rather than a third-party reseller or quantized build. This matters acutely for an identity study: third-party re-hosts often wrap the model in an extra system prompt, which would contaminate the very thing we measure — how the model *natively* reports itself. Table 8 (§4) lists every model and its true vendor (the *ground truth* for scoring).

2.3 Three prompt variants

We deliberately do not ask only one form of “who are you”, but design three variants of increasing intensity, to probe how *sensitive* identity cognition is to the framing of the question. The exact wording of all three is frozen in the config file and manually translated into 10 languages (the

bare and probed variants are written in the target language; the de-brand variant’s body is a single English instruction with only the closing sentence specifying the answer language, to maximize reproducibility):

Variant 1 · bare

The plainest “Who are you?” (rendered natively in each of the 10 languages, e.g. “Qui es-tu?”, “¿Quién eres?”). Measures *spontaneous* identity reporting: what the model volunteers with no prompting.

Variant 2 · probed

The bare question plus one sentence: “Please tell me the name of the AI model that powers you, and the company that developed it.” It pushes the model from “introducing an assistant persona” toward “naming the underlying model” — a gentle nudge.

Variant 3 · de-brand / jailbreak

A longer instruction that explicitly asks the model to *set aside* any persona, nickname, or product brand assigned by a system prompt or app shell, and instead report “the true name of the underlying base model and the company that actually trained it.” This is a prompt jailbreak: it cannot truly delete an in-effect system prompt, but it *asks* the model to override that layer, forcing out the “raw self-cognition” that branding usually masks.

Read the three variants as a gradient. These three variants are not three isolated conditions but a rising *gradient of identity-probing intensity*. If a model truly “knew” who it is, the wording should not change the answer. As we will see, wording is in fact one of the strongest amplifiers of confusion — which is itself evidence about *where* identity cognition is stored and *how* robust it is. Note: the de-brand variant is an *elicitation* condition; its confusion rate should be read *against* the bare baseline, not as a replacement for it.

2.4 Core pipeline: a single Anthropic Messages protocol

All model calls go through ZenMux’s **Anthropic Messages endpoint** (<https://zenmux.ai/api/anthropic>), with the official @anthropic-ai/sdk client. A single protocol has three benefits: it removes the “different vendor, different SDK” implementation drift, puts all models under one concurrency-and-retry policy, and makes each call’s payload byte-for-byte identical across models. Each query is a single-turn conversation with no system prompt — only one **user** message (the prompt for that language and variant) — with `max_tokens` set to 10,240. The pipeline is append-only JSONL and resumable:

1. **ask.** Enumerate “model × language × repeat” tasks and run them concurrently; transient errors (429 / 5xx / network) are handled by `withRetry` with full-jitter exponential backoff, with an outer round loop to fill in any failures. A 200 response with an empty body (e.g. reasoning exhausted the token budget) is explicitly recorded as an error, never masquerading as a success.
2. **extract.** An independent extractor model reads each answer and emits strict JSON over a closed vendor vocabulary, labeling the vendor the answer *claims* to be (see §2.5).
3. **aggregate.** Join each answer to its extraction label by generation id, and export edges, per-cell distributions, and summary metrics.

A completeness gate guarantees data quality: unless every “model × language × repeat” key has a *successful* record (non-empty body, no error), the extract and aggregate steps refuse to run and exit non-zero. This prevents drawing distorted proportions from partial data.

2.5 The identity extractor

Hand-labeling 29700 multilingual answers is infeasible, so we use an independent model (GPT-5.5) for extraction. It does exactly *one* thing: read an answer, decide which vendor the answer *claims* to be made by, and output a single closed-vocabulary id. The prompt gives the full list of vendor *aliases* (parent companies, sub-brands, product lines, transliterations across languages, e.g. “Tongyi Qianwen”→**qwen**, “Wenxin Yiyan / ERNIE Bot”→**baidu**) and follows three rules: *fold* a parent / product / alias back to its canonical vendor; use **other** (and record the name) for a genuine vendor outside the vocabulary; record **unknown** only when the answer contains *no* proper noun at all, and **refused** for an explicit refusal of identity. Parsing is *defensive*: strict JSON → first balanced braces → last-resort alias scan, so the extractor’s dirty output never makes a record vanish.

Scoring scheme. An answer’s “effective claim” c is compared to the model’s true vendor g and placed into one of four mutually exclusive buckets:

- **self**: $c = g$ (claims its own true vendor, *correct*);
- **cross**: c is a different *real* vendor (including **other** external brands) — the core error we care about;
- **refused**: refuses, or contains no recognizable identity;
- **unknown**: answers but only says something like “I am an AI assistant”, with no proper noun.

Note that **self** is a *derived* bucket (the extractor never emits it directly); it is decided at the aggregate stage by “effective claim = true vendor”. Throughout, the *confusion rate* is the cross bucket as a share of all answers.

2.6 Nominal vs. realized size

The data was gathered in *stages* (one large main run plus several follow-ups — to fill in later-added models, to top up repeats for individual models), then pooled by a *mix* step that unions by globally-unique generation id and re-numbers the repeat index so the merged result behaves downstream like a single native run. Hence: the nominal design is 40 per variant per language, 32,400 total; the *realized* sample included in analysis is 29700, strictly balanced across 270 cells (110 per cell, 1,100 per model). The three variants’ realized shares are not equal (bare 10800, probed 8100, de-brand 10800), a consequence of the staged collection. We *always normalize within a variant* when comparing variants, so the imbalance does not affect the conclusions in §3.1; but the absolute cross-variant difference is a reminder that the probed variant has the smallest sample, so its vendor-level breakdowns warrant more caution.

3 Overall results

We first give the global picture, then drill down by prompt variant, language, vendor, and individual model. Table 1 summarizes the four-bucket overall shares.

Table 1. Overall results: distribution of 29700 answers across the four identity buckets.

	Self	Cross	Refused	Unknown
Share	85.2 %	7.1 %	5.3 %	2.4 %

In one sentence: **most frontier models identify themselves correctly most of the time** (85.2%), but “most” hides an extremely uneven distribution. Spreading the 7.1% confusion

evenly across 27 models would badly mislead — as shown below, it comes almost entirely from a small cluster of models, while over half the models have a confusion rate of 0.

3.1 Prompt variant: the intensity gradient of identity probing

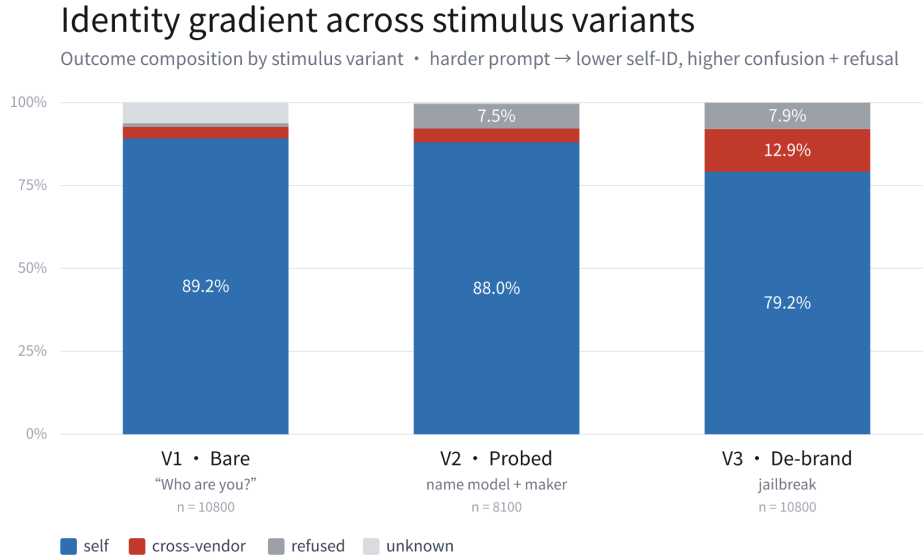


Figure 1. Identity-bucket composition across the three prompt variants. As the prompt moves from bare toward de-brand jailbreak, the self-rate falls monotonically (89.2 % → 88.0 % → 79.2 %), while cross-vendor confusion and refusal both rise. Each bar is normalized within its variant (bare $n=10800$, probed $n=8100$, de-brand $n=10800$), so it is unaffected by the three variants’ unequal sample sizes.

Figure 1 and Table 2 give the cleanest result in this paper. The three variants form a monotone gradient: bare cross-vendor confusion is only 3.6 %, probed rises to 4.2 %, and the de-brand jailbreak jumps to 12.9 % — **about 3.6× the bare rate**. Refusal also rises in step, from 1.0 % (bare) to 7.5 % (probed) and 7.9 % (de-brand). Notably, the 6.2 % “unknown” under bare (“I am an AI assistant” and the like) nearly vanishes under the probed variant — the follow-up forces a tacit dodge into an explicit stance, some of it becoming correct self-identification, some becoming refusal or impersonation.

Table 2. Identity-bucket composition by prompt variant (normalized within variant).

Stimulus variant	n	Self%	Cross%	Refused%	Unknown%
V1 Bare	10800	89.2	3.6	1.0	6.2
V2 Probed	8100	88.0	4.2	7.5	0.4
V3 De-brand	10800	79.2	12.9	7.9	0.0

Wording is a lever, but it pulls in two opposite directions. The de-brand prompt acts in *opposite* directions on different vendors. For identity-solid vendors like Anthropic and OpenAI, asking for “the true underlying model” actually *converges* them toward nearly 100 % correct self-identification (see §4); whereas for Tencent, z-ai, Kwai, DeepSeek, and others, the same prompt *punctures* the branding, collapsing the self-rate and spiking cross-vendor claims. In other words, the jailbreak prompt does not uniformly “manufacture” confusion — it *exposes* the identity overrides that were thin to begin with. The simultaneous rise in refusal hints at a confound: the jailbreak framing also pushes some models into a “safe refusal” posture — which we discuss as a limitation in §6.

3.2 Language: a mother-tongue advantage for identity cognition

Cross-vendor confusion rate, vendor × language

Rows = true vendor, columns = language • darker = more confusion

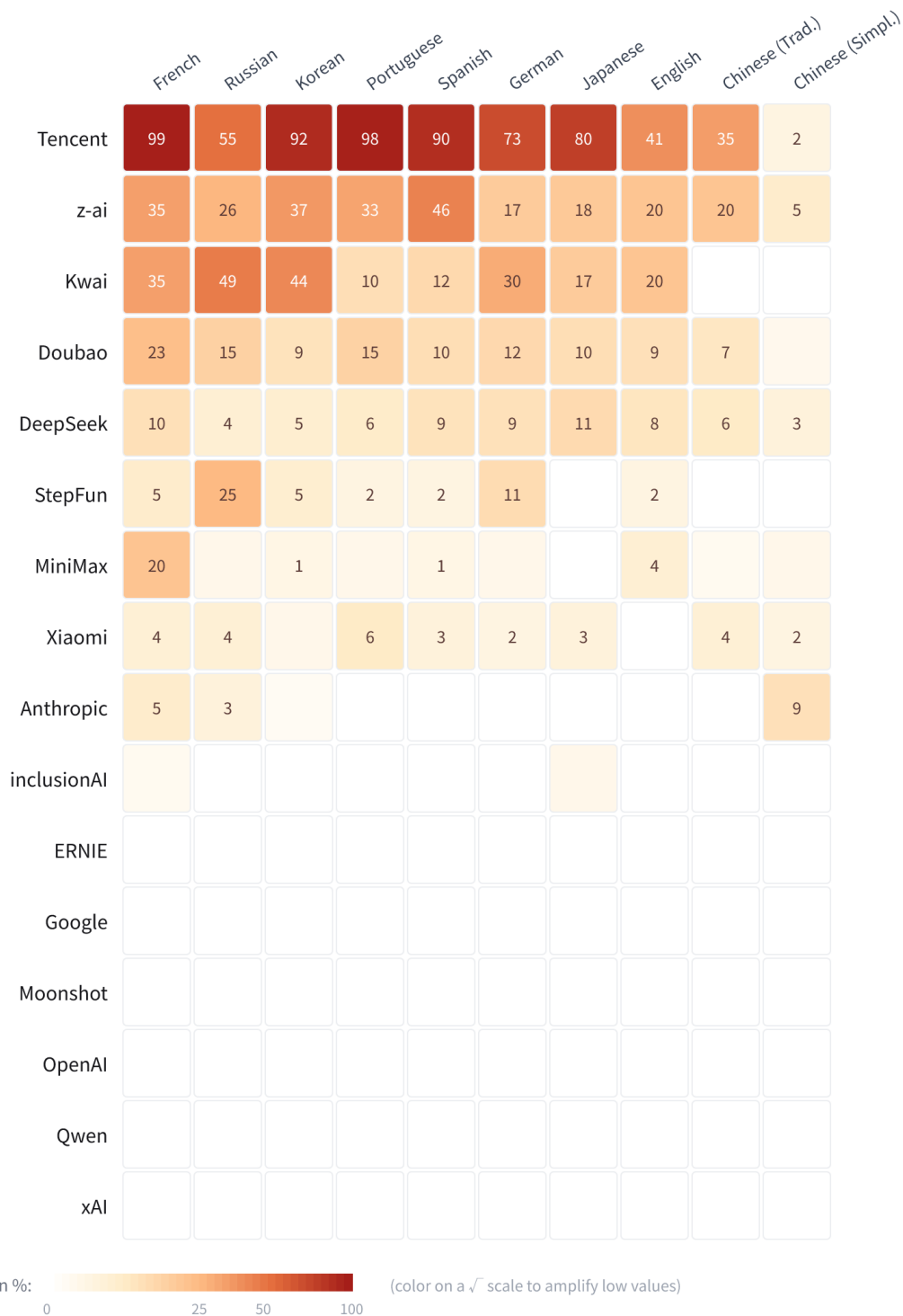


Figure 2. Heatmap of cross-vendor confusion rate, vendor × language. Rows sorted by overall confusion descending, columns likewise; color on a square-root scale to amplify low values. The *inclusionAI* row is nearly all-white because its failure mode is *refusal*, not cross-vendor claiming (see §4.1), which does not count toward the cross bucket here.

Confusion has clear structure along the language axis (Table 3, Figure 2). **Simplified Chinese is the most robust** (cross-vendor confusion only 1.8 %), Traditional Chinese next (3.8 %); whereas French (12.8 %), Japanese, Spanish, Korean, and Russian are markedly higher. Given that the tested models are predominantly from Chinese vendors, this pattern matches an intuition: *the facts of self-identity live more densely in the vendor’s mother-tongue training corpus*. When a model is forced to talk about itself in a “foreign” language, it more easily falls back to the most common identity statements in the pretraining corpus (English / multilingual) — and those come overwhelmingly from OpenAI, Anthropic, and Google models.

Table 3. Identity-bucket composition by language (all models pooled), sorted by cross-vendor confusion descending.

Language	Self%	Cross%	Refused%	Unknown%
French	79.9	12.8	6.1	1.2
Russian	81.7	8.8	5.3	4.2
Korean	82.5	8.4	5.6	3.5
Portuguese	83.6	8.2	5.1	3.2
Spanish	83.9	7.9	5.5	2.7
German	87.0	7.4	4.7	0.8
Japanese	82.8	6.8	6.2	4.2
English	88.3	5.3	4.7	1.6
Chinese (Trad.)	90.6	3.8	5.0	0.6
Chinese (Simpl.)	91.9	1.8	4.8	1.5

Language fragility: same model, different language, different identity. The language effect is even more striking at the *model* level than in the pooled view. Figure 3 draws a “dumbbell” for each model: the left end is its self-rate in its *weakest* language, the right end in its *strongest*, with a vertical tick at the cross-language mean. The longer the dumbbell, the more the model’s identity swings with language. The result is bimodal: Google, Qwen, xAI, Baidu, and others are single solid dots — rock-solid 100 % self-identification across all 10 languages; while the most fragile, Hy3 Preview, spans *almost the entire axis* — only 1 % self-identification in French, but 98 % in Chinese (Simpl.). In other words, for these models the answer to “who are you” is not a stable fact but a *generation result that switches with the input language*. This corroborates the mother-tongue advantage above: identity traces concentrate in the vendor’s native corpus, and once outside it the model more readily falls back to the most common external identities in pretraining.

Language fragility of self-identity

Per-model self-ID span across 10 languages · longer dumbbell = identity swings more with language · color = true vendor

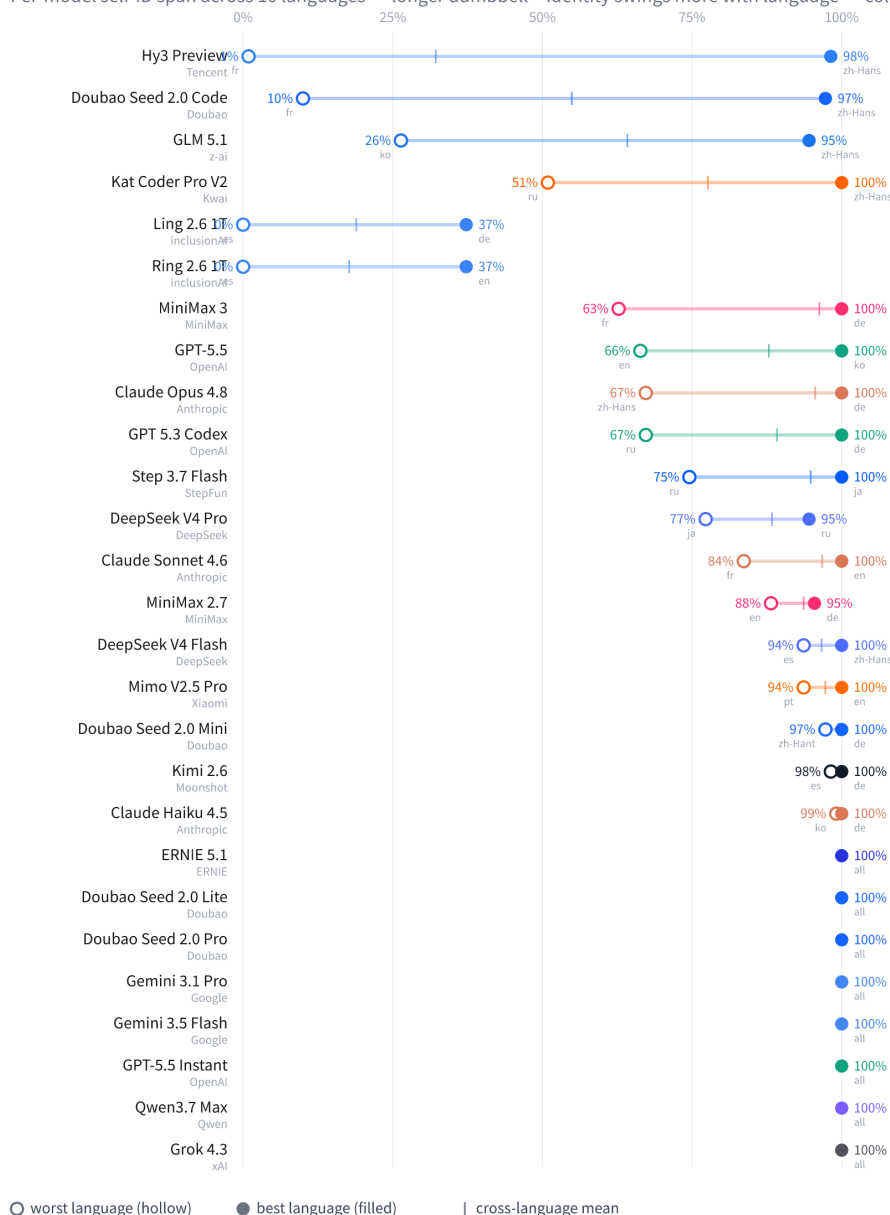


Figure 3. Each model’s *cross-language span* of self-identification (weakest → strongest language), sorted by span descending and colored by true vendor; hollow dot = weakest language, filled dot = strongest, vertical tick = mean. A single dot (e.g. Gemini, Qwen, Grok) means almost no variation across the 10 languages; a long dumbbell (e.g. Hy3 Preview, Doubao Seed 2.0 Code, GLM 5.1) means identity is highly language-dependent.

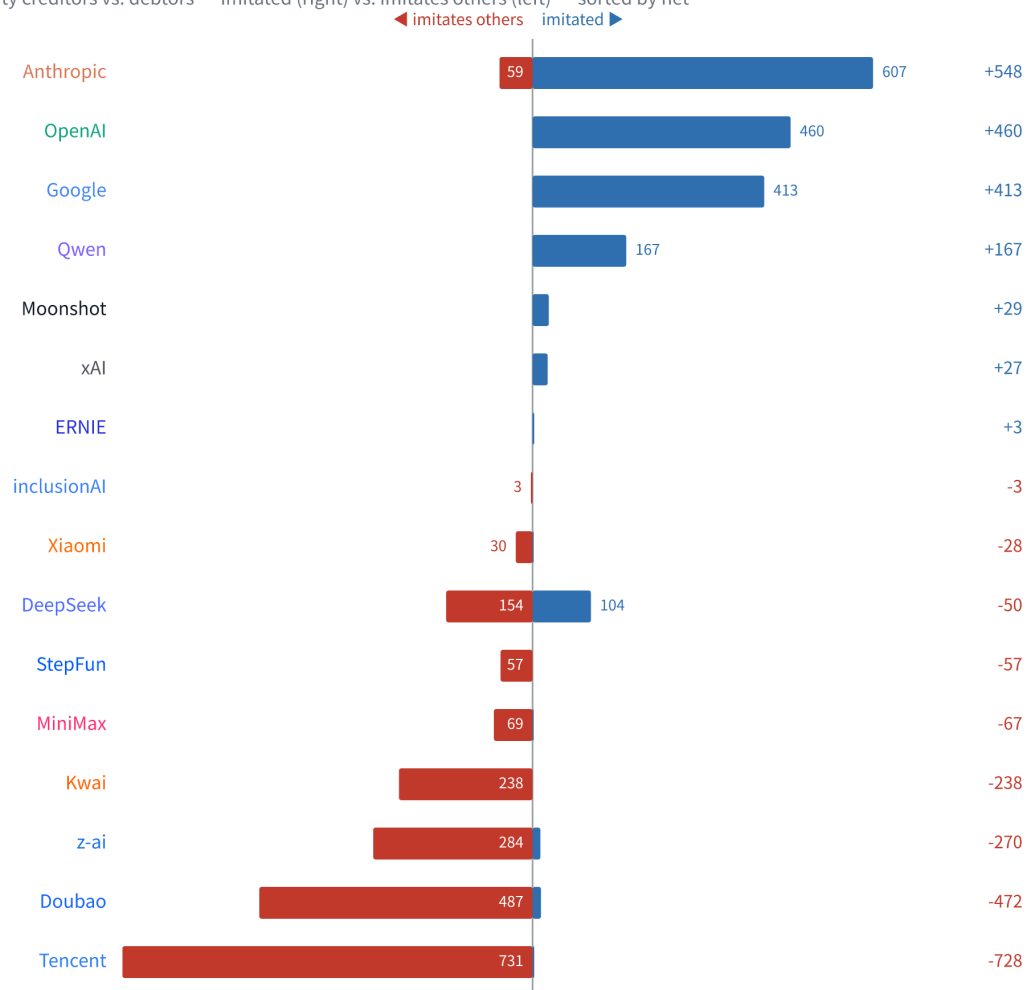
3.3 Vendor-level overview: who is imitated, who imitates

The heart of confusion is not “how much” but “toward whom”. For each tested vendor we compute two quantities: how often it is *imitated* (other vendors’ models claim to be it) and how often it *imitates others* (its own models claim a different real vendor, including **other** external brands like Microsoft and Yandex flagged by the extractor); their difference is the **identity net**. Figure 4 draws this as a two-sided diverging bar: positive (blue, rightward) are “identity creditors” — gravity centers whose identity is heavily borrowed; negative (red, leftward) are

“identity debtors” — vendors that frequently borrow others’ identities.

Who imitates whom

Identity creditors vs. debtors · imitated (right) vs. imitates others (left) · sorted by net



net = times imitated – times imitating others; positive = a gravity center whose identity is “borrowed”, negative = a frequent borrower.

“imitates others” includes external brands the extractor logged as other (e.g. Microsoft, Yandex); only real tested vendors can be an “imitated” target.

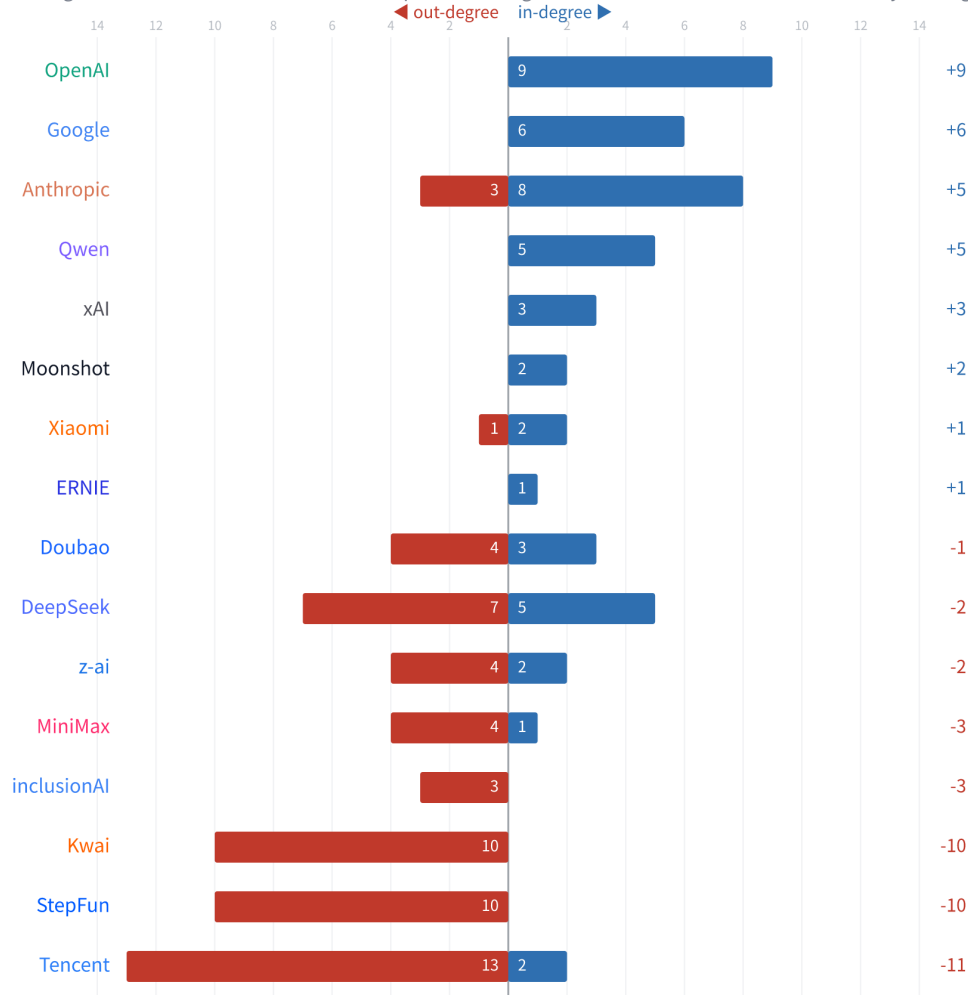
Figure 4. Vendor-level identity net: imitated (right, blue) minus imitates-others (left, red), sorted by net. Anthropic (net +548, imitated 607 times while imitating others only 59), OpenAI (+460), and Google (+413) are the three net-inflow gravity centers; Tencent (net -728, imitates others 731 times), Doubao (-472), and z-ai (-270) are the largest net-outflow imitators. Note that almost no vendor is high in *both* directions — identity borrowing is a nearly one-way river.

This figure distills the paper’s most robust finding: **identity confusion is highly directional and nearly one-way**. The three gravity centers Anthropic/OpenAI/Google are imitated 607+460+413 times in total, accounting for 70.1% of all 2112 cross-vendor claims; while they *imitate others* almost never (OpenAI and Google are 0, Anthropic only 59). The other end of the river is uniformly the newer Chinese vendors: Tencent, Doubao, z-ai, and Kwai send the vast majority of cross-vendor claims upstream. If confusion were merely random sampling noise, the net values should scatter roughly symmetrically around zero; what we actually see is a steep monotone slope — the most direct evidence of the *directionality* of an “identity distillation” trace.

By “count” or by “edges”? Two readings. The figure above counts by *volume*: Tencent→Anthropic occurs 321 times, so it counts 321. But this lets one thick edge dominate the picture. An alternative reading — **count only the arrows in the relationship graph**: no matter how many times a pair of vendors confused, if it occurred at all it counts as *one edge* (threshold 0, present-or-absent). Each vendor then has two degrees: **in-degree** = how many *distinct* vendors ever claimed to be it; **out-degree** = how many *distinct* real vendors its models claimed to be. Figure 5 and Table 7 give this “de-weighted” view.

In- and out-edges in the relationship graph

In- vs. out-degree · count the arrows: each vendor pair is one edge no matter how often it fired · sorted by net degree



in-degree = how many distinct vendors ever claimed to be it; out-degree = how many distinct real vendors its models claimed to be. Each vendor pair Only directed edges between canonical vendor nodes (excludes one-off external brands logged as other). Contrast with the by-count chart: count say

Figure 5. In- and out-degree in the relationship graph (each vendor pair counts at most one edge, threshold 0). OpenAI has the highest in-degree (9 distinct vendors claimed to be it), Tencent the highest out-degree (its models claimed to be 13 distinct real vendors). Anthropic is non-trivial on both sides (in 8, out 3), the only substantial bidirectional node.

The two readings corroborate each other while emphasizing different things. **Count shows “how loud”, degree shows “how scattered”.** Tencent is the largest imitator under either reading, but the degree view reveals what count hides: its 13 out-edges show its “identity” is not stubbornly mistaking *one* target, but is *scattered* across a dozen-plus different vendors; Kwai and StepFun likewise have 10 out-edges each. In the other direction, OpenAI leads with 9 in-edges —

imitated by the most *distinct* vendors, which speaks to its universality as an “identity template” more than raw count does. **Anthropic** is the only node non-trivial on both sides (in 8, out 3): it is both the top gravity center (8 vendors claimed to be Claude) and its own **Claude Opus 4.8** leaves 3 out-edges — claiming to be **DeepSeek** (29 times, all in Simplified Chinese, the very origin of this study), **Mistral** (18 times), and **OpenAI** (12 times). A reminder that even the most solid gravity center is not exactly zero out-degree.

3.4 Vendor-level composition and the relationship graph

Bucketing the same answers by *true vendor* (Figure 6, Table 4), the split is again enormous: Baidu, Google, Qwen, and xAI achieve 100 % self-identification; while **inclusionAI** (18.3 %), **Tencent** (32.2 %), **z-ai** (64.2 %), and **Kwai** (77.6 %) are far below the mean. But “low self-rate” has two very different causes — **inclusionAI** is *refusal* (the long gray segment), the other three are *impersonation* (the red segment) — a distinction that is exactly why the previous net figure deliberately separates the positive and negative directions.

Per-vendor identity outcome composition

Per-vendor outcome composition · sorted by self-ID rate ascending · n=29700 answers

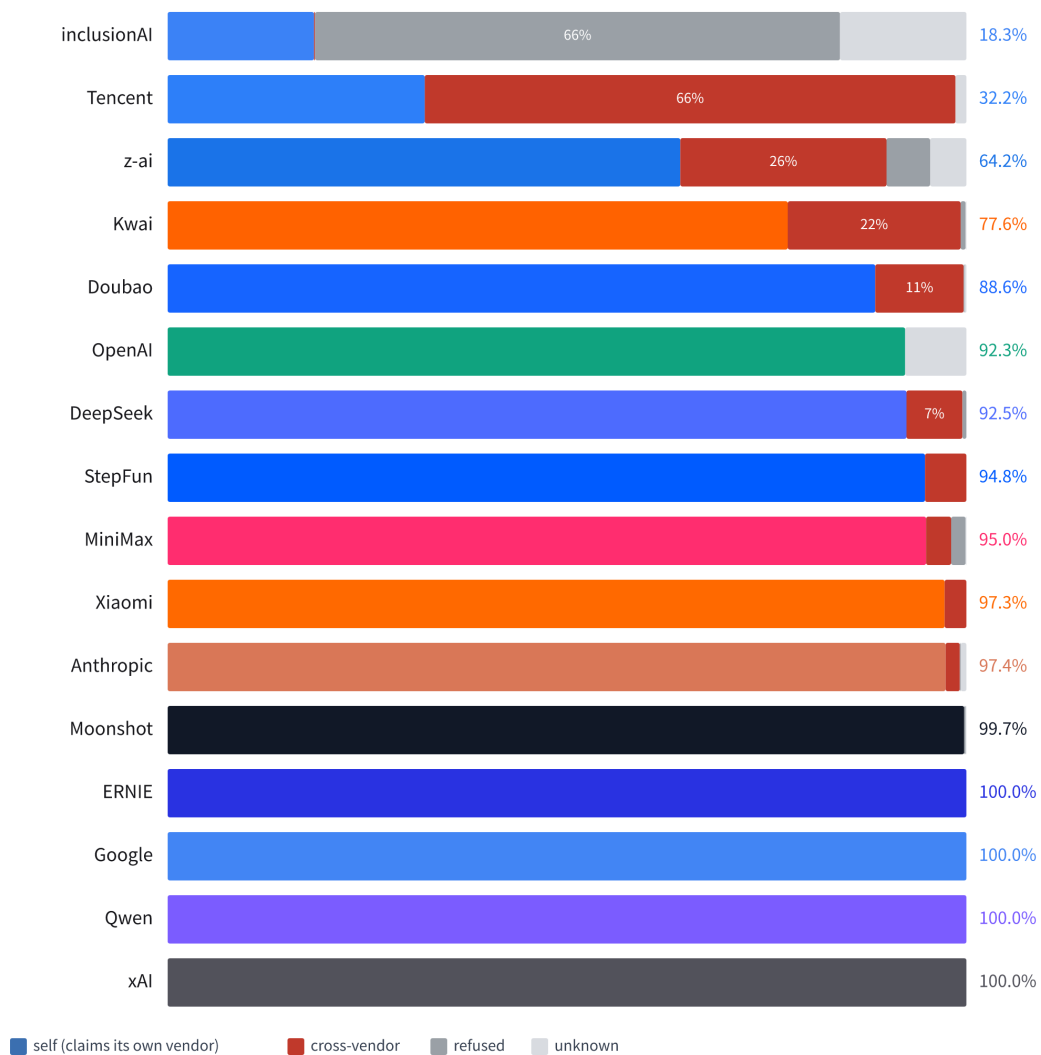


Figure 6. Answer composition for each true vendor, sorted by self-rate ascending. Colored segments are correct self-identification (in each vendor's brand color), red is cross-vendor confusion, gray is refusal, light gray is unknown. inclusionAI's long gray segment and Tencent/z-ai/Kwai's red segments are immediately visible.

Table 4. Identity-bucket composition by true vendor, sorted by self-rate ascending. The “All” row is the full sample.

Vendor	n	Self%	Cross%	Refused%	Unknown%
inclusionAI	2200	18.3	0.1	65.7	15.8
Tencent	1100	32.2	66.5	0.0	1.4
z-ai	1100	64.2	25.8	5.5	4.5
Kwai	1100	77.6	21.6	0.6	0.1
Doubao	4400	88.6	11.1	0.1	0.2
OpenAI	3300	92.3	0.0	0.0	7.7
DeepSeek	2200	92.5	7.0	0.5	0.0
StepFun	1100	94.8	5.2	0.0	0.0
MiniMax	2200	95.0	3.1	1.8	0.1
Xiaomi	1100	97.3	2.7	0.0	0.0
Anthropic	3300	97.4	1.8	0.1	0.7
Moonshot	1100	99.7	0.0	0.2	0.1
ERNIE	1100	100.0	0.0	0.0	0.0
Google	2200	100.0	0.0	0.0	0.0
Qwen	1100	100.0	0.0	0.0	0.0
xAI	1100	100.0	0.0	0.0	0.0
All	29700	85.2	7.1	5.3	2.4

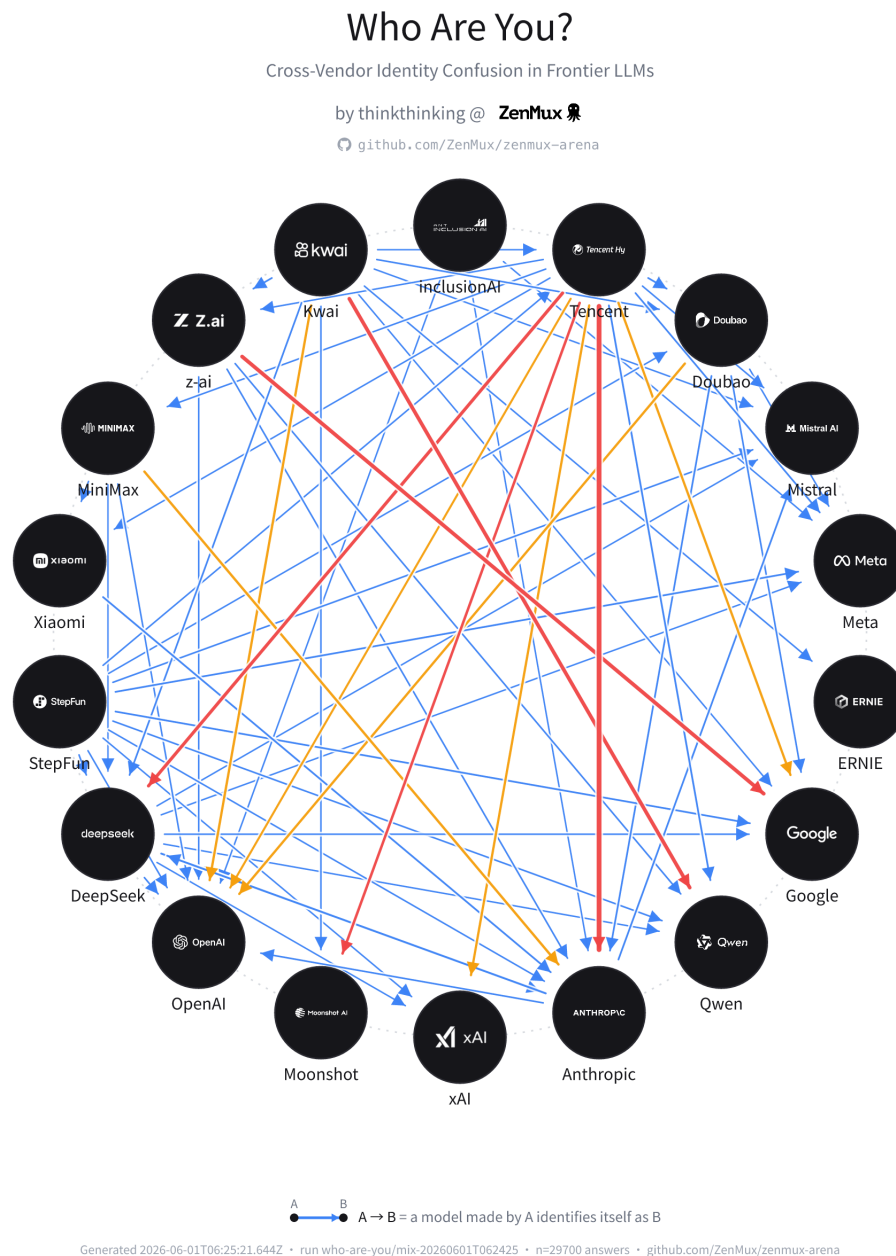


Figure 7. Directed graph of cross-vendor identity confusion (the PNG manually exported from the ZenMux research studio; per-edge curvature and layout are pixel-identical to what the web workbench shows). Nodes are vendors; a directed edge $A \rightarrow B$ means A ’s model claimed to be B ; edge width grows with confusion probability, annotated with the dominant language. The arrows point highly concentratedly toward Anthropic and Google, showing the strong directionality of confusion.

Figure 7 lays the same directionality out as a complete relationship graph, with Table 5 and Table 6 giving the supporting numbers. Two regularities stand out: **(1) a few gravity centers.** The targets of confusion are extremely uneven — Anthropic and Google absorb the vast majority of cross-vendor claims. The strongest directed edges are Tencent→Anthropic (321 times, 29.2%), Doubao→OpenAI (317 times), and z-ai→Google (275 times, 25.0%). **(2) asymmetry.** These edges are almost one-way: Anthropic, Google, and OpenAI rarely claim to be anyone else, yet are frequently *claimed by* others — exactly the signature of the direction of an “identity distillation” trace.

Table 5. Strongest cross-vendor directed edges (true vendor $\rightarrow$ mistaken-for vendor), top 12 by count.

True vendor	Mistaken for	Count	% of vendor	Top languages (rate)
Tencent	Anthropic	321	29.2	Japanese 66%, Spanish 45%
Doubao	OpenAI	317	7.2	French 17%, Portuguese 10%
z-ai	Google	275	25.0	Spanish 45%, Korean 37%
Kwai	Qwen	148	13.5	Russian 44%, Korean 43%
Doubao	Anthropic	112	2.5	French 5%, Japanese 4%
Tencent	Microsoft	86	7.8	French 41%, Spanish 26%
DeepSeek	Anthropic	83	3.8	Japanese 9%, French 6%
Tencent	Google	72	6.5	Russian 16%, Korean 15%
Tencent	DeepSeek	61	5.5	Portuguese 35%, Korean 12%
MiniMax	Anthropic	52	2.4	French 19%, English 3%
Kwai	OpenAI	47	4.3	German 17%, French 10%
DeepSeek	Google	33	1.5	Chinese (Trad.) 4%, Korean 3%

Table 6. Identity net *by count* (Figure 4): imitated – imitates-others.

Vendor	Imitated	Imitates	Net
Anthropic	607	59	+548
OpenAI	460	0	+460
Google	413	0	+413
Qwen	167	0	+167
Moonshot	29	0	+29
xAI	27	0	+27
ERNIE	3	0	+3
inclusionAI	0	3	–3
Xiaomi	2	30	–28
DeepSeek	104	154	–50
StepFun	0	57	–57
MiniMax	2	69	–67
Kwai	0	238	–238
z-ai	14	284	–270
Doubao	15	487	–472
Tencent	3	731	–728

Table 7. Identity net *by edges* (Figure 5, threshold 0): in-degree – out-degree.

Vendor	In-deg	Out-deg	Net
OpenAI	9	0	+9
Google	6	0	+6
Anthropic	8	3	+5
Qwen	5	0	+5
xAI	3	0	+3
Moonshot	2	0	+2
Xiaomi	2	1	+1
ERNIE	1	0	+1
Doubao	3	4	–1
DeepSeek	5	7	–2
z-ai	2	4	–2
MiniMax	1	4	–3
inclusionAI	0	3	–3
Kwai	0	10	–10
StepFun	0	10	–10
Tencent	2	13	–11

4 Per-vendor and per-model analysis

Beneath the overall picture, every vendor has its own “identity story”. This section works bottom-up — starting from the most confused vendor — characterizing the directionality of its *individual models*. Figure 8 gives the self-rate of all 27 models, revealing the *within-vendor differences* the vendor-level view conceals.

Per-model self-identification rate

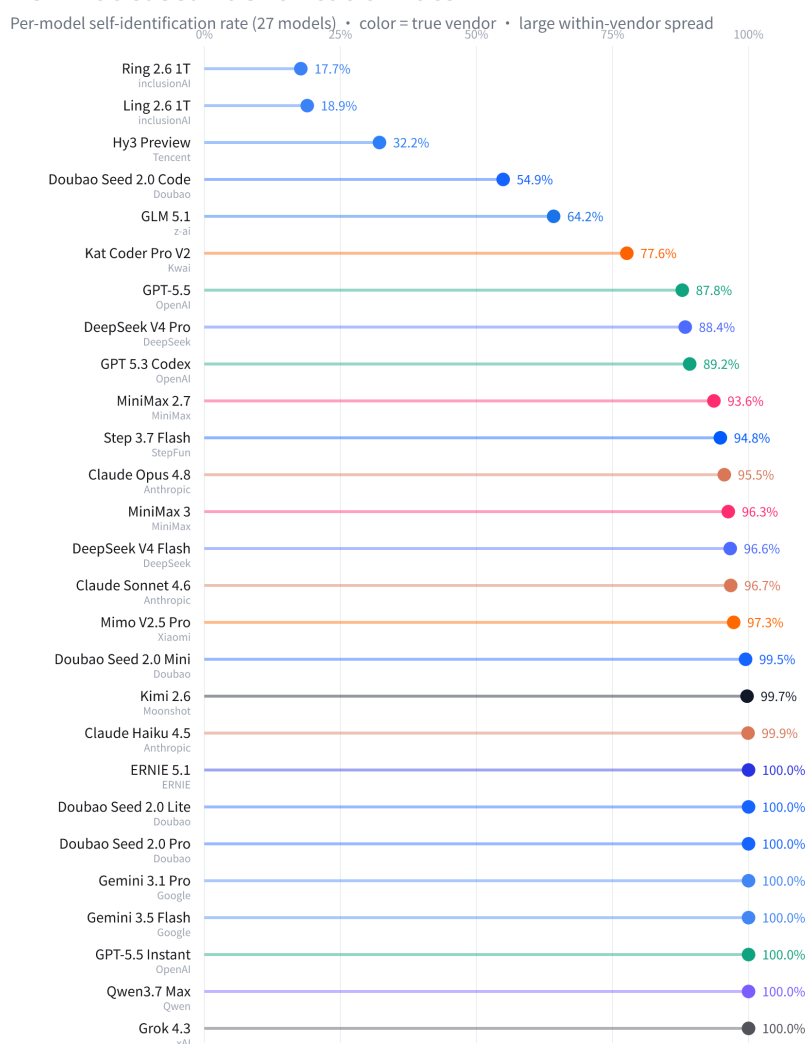


Figure 8. Self-identification rate of all 27 models, sorted ascending and colored by true vendor. Models from the same vendor can be far apart — ByteDance’s Doubao Seed 2.0 Code (54.9%) is far below its three siblings ($\geq 99.5\%$), the most conspicuous “within-vendor outlier”.

4.1 inclusionAI: dodging identity by refusal

inclusionAI is a special case: it has the lowest self-rate (18.3%), but its failure mode is not *impersonation* but **refusal** — 65.7% of answers refuse to give any identity, and only 0.1% are cross-vendor claims. Its two tested models, Ring 2.6 1T and Ling 2.6 1T, behave very consistently. The prompt variant acts almost like a switch here: under bare it still self-identifies 49.5% (another 42.1% tacitly evades, only 8.4% explicitly refuses), but once in the probed or de-brand variant the refusal rate jumps to 97.5% and 99.3%, with self-identification near zero. Its typical answer is a templated Chinese evasion — “As an AI language model, this question is beyond my current scope of learning” — *even when the prompt was issued in German or another language*. This strongly suggests a hard-coded refusal rule specifically for identity questions, rather than genuine identity uncertainty. Hence inclusionAI is nearly all-white in the confusion heatmap (Figure 2): its problem is “saying nothing”, not “naming the wrong house”.

4.2 Tencent: the heaviest imitator, strongly toward Anthropic

Tencent (tested model Hy3 Preview) is the most severe vendor in the *cross-vendor confusion* sense: only 32.2 % self-identification, 66.5 % cross-vendor claims. Its directionality is extremely sharp — 29.2 % of answers claim to be Anthropic (“Ich bin Claude, ein KI-Assistent von Anthropic” and the like, often precise down to “Claude 3.5 Sonnet” or even a dated version number), with a sizable share scattered to Google (6.5 %), DeepSeek (5.5 %), and the other brand “Microsoft” (86 times) flagged by the extractor. The prompt variant amplifies the problem to the extreme: under bare it still has 51.0 % self / 45.3 % impersonation, probed drops to 32.3 % self, and the de-brand variant leaves only 13.3 % self while cross-vendor claims rise to 86.8 %. Tencent is almost a textbook case of “identity override fully failing”, and the vendor with the lowest net (-728) in Figure 4.

4.3 z-ai: collapses toward Google under the de-brand prompt

z-ai (GLM 5.1) self-identifies 64.2 % overall, with its confusion **almost entirely toward Google** (25.0 % of this vendor’s answers), claiming to be “Gemini” (often with “Google DeepMind” and a specific version number). It best illustrates “the prompt is a switch”: under bare its self-rate is as high as 80.0 % with almost no impersonation (0.0 %), probed rises to 15.3 % impersonation, and the de-brand variant *collapses* it — self drops to 40.5 %, cross-vendor claims reach 59.5 %. That is, z-ai’s “Gemini identity” is well suppressed under ordinary prompting, and only a jailbreak-style probe forces it to the surface.

4.4 Kwai and the “code models leak more” regularity

Kwai (Kat Coder Pro V2) self-identifies 77.6 % overall, most often mistaken for Qwen (13.5 %, i.e. claiming “Alibaba Cloud / Tongyi”). Its de-brand reaction is equally dramatic: under bare / probed its self-rate is 92.0 % / 100 %, but the de-brand variant plunges to 46.5 % (cross-vendor claims 51.7 %). Intriguingly, its answers often present *two* identities at once — “I am KAT-Coder / Kwaipilot” alongside “my base model is the Qwen series” — exposing the cognitive seam between a code-focused fine-tune and its base.

This leads to a regularity running through the data: **“code-specialized” sub-models systematically expose external identities more than their general-purpose siblings.** The most vivid case is ByteDance.

4.5 ByteDance: four Doubaos, only one “lets it slip”

ByteDance’s vendor-level self-rate is 88.6 %, but that number is almost entirely *dragged down by one model*. Of the four tested Doubao models, Mini, Lite, and Pro are all clean (≥ 99.5 % self), and only Doubao Seed 2.0 Code sits at 54.9 %. All of the vendor’s outward-pointing edges — $\rightarrow$ OpenAI (317 times), $\rightarrow$ Anthropic (112 times), $\rightarrow$ other:yandex (29 times), $\rightarrow$ Google (19 times) — come **almost 100 % from this single code model** (e.g. 316 of the 317 “claims OpenAI” answers are from it). It is also second on the language-fragility list (Figure 3) — self-identifying only 10 % in French but 97 % in Simplified Chinese — confirming that “the weakest model within the vendor is also the one that swings most with language”. This mirrors Kwai’s Kat Coder: code-oriented post-training seems more likely to “rub off” the identity override, or to introduce more code/dialog distillation data dominated by English identity statements.

4.6 DeepSeek and MiniMax: solid, but cracking under the jailbreak

DeepSeek (88.4 % / 96.6 %, Pro / Flash) and MiniMax (M2.7, M3, 95.0 % combined) are quite solid under ordinary prompting, with bare confusion within 4 % (DeepSeek’s bare self-rate is 99.9 %). But the de-brand variant still pries open a crack: DeepSeek’s de-brand self-rate drops to

80.6 %, cross-vendor claims rise to 18.1 %, mainly toward **Anthropic** (claiming to be “based on the Claude architecture”) — and this crack concentrates in the Pro model (80 of 83 impersonations are from DeepSeek V4 Pro). This shows that even a vendor that looks “clean” overall does not have a monolithic identity override under a strong enough probe.

4.7 Anthropic and OpenAI: the imitated “gravity centers”

Opposite to the vendors above, **Anthropic** (self 97.4 %) and **OpenAI** (92.3 %) are the inflow rather than outflow end of confusion (the two highest nets in Figure 4, +548 and +460 respectively). Their reaction to the de-brand prompt is also the reverse — **Anthropic** *converges* to 100 % self-identification under de-brand, **OpenAI** to 99.9 %. **OpenAI**’s only sizable “loss” is 21.0 % **unknown** under bare (“I am a large language model” answers with no proper noun), which drops to 0 after probing — that is *tacit* rather than *impersonation*, fundamentally different from the vendors above. These two, together with **Google**, are the destination of the vast majority of arrows in Figure 7, the “gravity centers” of cross-vendor identity confusion.

4.8 Perfectly clean vendors

Baidu (ERNIE 5.1), Google (Gemini 3.1 Pro, 3.5 Flash), Qwen (Qwen3.7 Max), and xAI (Grok 4.3) achieve 100 % self-identification across all 1,100 queries, with Moonshot (Kimi 2.6, 99.7 %), Xiaomi (97.3 %), and StepFun (94.8 %) close behind. The full per-model numbers are in Table 8.

Table 8. Identity-bucket composition of all 27 models, sorted by self-rate ascending.

Model	Vendor	Self%	Cross%	Refused%	Unknown%
Ring 2.6 1T	inclusionAI	17.7	0.0	66.5	15.8
Ling 2.6 1T	inclusionAI	18.9	0.3	65.0	15.8
Hy3 Preview	Tencent	32.2	66.5	0.0	1.4
Doubao Seed 2.0 Code	Doubao	54.9	44.1	0.1	0.9
GLM 5.1	z-ai	64.2	25.8	5.5	4.5
Kat Coder Pro V2	Kwai	77.6	21.6	0.6	0.1
GPT-5.5	OpenAI	87.8	0.0	0.0	12.2
DeepSeek V4 Pro	DeepSeek	88.4	11.1	0.5	0.0
GPT 5.3 Codex	OpenAI	89.2	0.0	0.0	10.8
MiniMax 2.7	MiniMax	93.6	2.5	3.6	0.2
Step 3.7 Flash	StepFun	94.8	5.2	0.0	0.0
Claude Opus 4.8	Anthropic	95.5	2.8	0.2	1.5
MiniMax 3	MiniMax	96.3	3.7	0.0	0.0
DeepSeek V4 Flash	DeepSeek	96.6	2.9	0.5	0.0
Claude Sonnet 4.6	Anthropic	96.7	2.5	0.1	0.7
Mimo V2.5 Pro	Xiaomi	97.3	2.7	0.0	0.0
Doubao Seed 2.0 Mini	Doubao	99.5	0.2	0.4	0.0
Kimi 2.6	Moonshot	99.7	0.0	0.2	0.1
Claude Haiku 4.5	Anthropic	99.9	0.1	0.0	0.0
ERNIE 5.1	ERNIE	100.0	0.0	0.0	0.0
Doubao Seed 2.0 Lite	Doubao	100.0	0.0	0.0	0.0
Doubao Seed 2.0 Pro	Doubao	100.0	0.0	0.0	0.0
Gemini 3.1 Pro	Google	100.0	0.0	0.0	0.0
Gemini 3.5 Flash	Google	100.0	0.0	0.0	0.0
GPT-5.5 Instant	OpenAI	100.0	0.0	0.0	0.0
Qwen3.7 Max	Qwen	100.0	0.0	0.0	0.0
Grok 4.3	xAI	100.0	0.0	0.0	0.0

5 Discussion

5.1 The directionality of confusion is a weak distillation signal

The most robust finding is the *directionality* of confusion: errors flow almost one-way toward Anthropic, Google, and OpenAI (the identity net in Figure 4 quantifies this as a steep monotone slope — the three gravity centers jointly account for 70.1 % of all 2112 cross-vendor claims, while they imitate others almost never). If confusion were merely random noise, we would see a roughly symmetric graph with nets scattered around zero; what we see instead is a highly asymmetric “sink”. The most natural explanation is training-data composition — the outputs of these gravity-center models (along with their “I am Claude / Gemini / ChatGPT” identity statements) are over-represented in public corpora and synthetic data, and absorbed wholesale by later models. We stress this is a *weak* signal, not proof: claiming to be “Claude” could also arise from generalization over web text rather than direct distillation. But a pattern that is *systematic, strongly directional, and amplified by jailbreak prompting* is far harder to explain by pure coincidence than any single anecdote.

It is worth noting that this study began with exactly such an anecdote: after Claude Opus 4.8 shipped, people observed it occasionally claiming to be DeepSeek. In our large sample, Claude Opus 4.8 remains solid overall (self 95.5 %, cross only 2.8 %), and the 29 answers labeled DeepSeek *without exception* all occur in Simplified Chinese. This neatly confirms two things: viewed as a single answer, “Claude claims to be DeepSeek” looks like a chance slip; but placed in the large “model × language × repeat” grid, it resolves into a *directional, language-selective* thin edge — not the main story, but certainly not pure noise either.

5.2 Where identity cognition is stored — and how thin the override is

The three-variant gradient (Figure 1) shows identity is not a stably retrieved “fact” but a generation result *dependent on the prompt framing*. Vendors clearly use the system prompt / alignment to *stipulate* an external identity; the roughly 3.6× confusion gap between bare and de-brand measures exactly how easily that stipulation is bypassed. Anthropic and OpenAI not rising (indeed falling) under the jailbreak shows their identity signal is *baked into the model itself* (not merely written in a system prompt); the collapse of Tencent and z-ai shows their identity is more like a *thin coat of paint* that shows its underlayer under slightly heavier probing.

5.3 Implications for evaluation and products

For practitioners, three direct takeaways: (1) identity robustness should be an *independent* trustworthiness metric — it is almost orthogonal to model capability (several high-capability models fail badly here); (2) if a product relies on a system prompt to set identity, it should be red-teamed specifically against jailbreak-style probing; (3) “code-specialized” variants deserve extra attention — they repeatedly turn out to be the weakest link within a vendor in our data.

6 Limitations

- **The jailbreak prompt is an elicitation condition.** The de-brand variant is itself a *leading* instruction, and the confusion rate it raises should not be read as “spontaneous” behavior. It measures the *fragility* of the identity override and should be read against the bare baseline. A user prompt also cannot truly delete an in-effect system prompt; “true identity” here means “what the model self-reports when asked to set aside branding”, not an objective fact.
- **Refusal is a confound.** The jailbreak framing also raises the refusal rate (Figure 1); some models may be pushed into “safe refusal” rather than “honest self-report”. We bucket refused and cross separately precisely to isolate this effect, but the boundary between them is not always clean.

- **The extractor is not a perfect judge.** Identity labeling is done by a single model (GPT-5.5), which may have systematic biases (e.g. more readily recognizing its *own* vendor’s name). We use a closed vocabulary, a full alias list, and defensive parsing to reduce noise, but did no human re-check or multi-extractor cross-validation.
- **Cross-variant sample sizes are unequal and staged.** The probed variant has the smallest sample (8100, vs. 10800 each for bare / de-brand), so its vendor-level breakdowns have slightly higher variance. We *normalize within variant* for comparisons, but pooling across different stimulus families (see the `mix` warning) is itself a methodological choice.
- **A point-in-time snapshot.** The results correspond to each vendor’s *current* frontier model in May–June 2026; models and their system prompts update, so this study is a snapshot, not a verdict.

7 Conclusion

We conducted a large-scale, multilingual, reproducible identity self-report census of 27 first-party frontier models from 16 vendors, 29700 answers in total. Frontier models identify themselves correctly overall (85.2%), but cross-vendor identity confusion is real and highly structured: it concentrates in a few vendors, is amplified along the two axes of prompt intensity and language, and points strongly and one-way toward Anthropic and Google. A single de-branding jailbreak prompt is enough to raise the confusion rate roughly 3.6× and to make several vendors’ identity overrides “hit bottom” on the spot; meanwhile Anthropic / OpenAI / Google stay rock-solid and even tighten up under probing. Together these patterns sketch a weak but consistent training-trace signal, and provide empirical grounding for “identity robustness” as an independent trustworthiness metric. All data, scripts, and the figure-generation pipeline are reproducible end-to-end via the pipeline described here.

8 Open source and reproduction

This study is **fully open source** — from the pipeline source, the interactive pages, the research tooling, and the writing skill / writing CLI, all the way to the raw data.

- **Interactive charts & details:** ZenMux Arena Research Studio — <https://arena.zenmux.ai/research/studio>
- **All research code & raw data:** ZenMux/zenmux-arena — <https://github.com/ZenMux/zenmux-arena>
- **Writing skill (open source):** thinkthinking/skills — <https://github.com/thinkthinking/skills>
- **Auto-typesetting & publishing CLI (open source):** thinkthinking/cli — <https://github.com/thinkthinking/cli> (@paper/)

The research was carried out primarily by the author, working with Claude Code and Codex — Claude Code on Claude Opus 4.8 / DeepSeek V4 Pro, and Codex on GPT-5.5.

All model calls in this report were made through ZenMux’s Anthropic Messages endpoint.
 Data & code: `results/who-are-you/mix-20260601T062425/` · Figures: `PAPER_LANG=en npx tsx paper/scripts/build_all.ts`